

Vertical Non-Injectivity and the Stability of the Observable Rank: Closing the Unconditional Transfer $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$

O24 — Spectral Admissibility Sub-programme

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Abstract

O18 [26] established the minimal fibre condition: Born–Infeld parity is the only global symmetry of $S_{\text{BI}}[\chi]$, ensuring that each fibre of the non-injective projection Π is exactly $\{\chi, -\chi\}$. O22 [30] proved the shell-alignment conjecture of O21 [29], and O23 [31] derived the threshold condition $\Sigma_c(n_3) = 3$ from the quaternionic maximality of the admissible neutral traceless sector. Together, these results show that the integer 3 in $\Sigma_c(n_3) = 3$ is a property of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$, not of the cardinality of the fibres. The condition of O18 is therefore stronger than necessary: the programme requires only that any symmetry of S_{BI} acts *vertically* with respect to Π , i.e. does not enlarge the real rank of the admissible neutral traceless sector. The present paper proves this *verticality condition* directly from Born–Infeld admissibility and the quaternionic maximality established in O23, without requiring a classification of $\mathcal{C}_{\text{eff}}$. As a corollary, the transfer chain

$$c_\chi \longrightarrow \delta_{\text{pair}} \longrightarrow \beta^* = \frac{1}{\delta_{\text{pair}} + \frac{1}{2}} \approx 0.126$$

is established unconditionally with respect to the fibre structure of Π . Non-injectivity may increase microscopic multiplicity, but cannot enlarge the admissible observable structure.

Keywords. Cosmochrony; spectral admissibility; non-injective projection; verticality condition; observable rank stability; quaternionic maximality; Born–Infeld constraint; fibre structure; cascade exponent β^*

Contents

1	Position of the Problem after O22–O23	3
1.1	What O18 required and why it was sufficient but not necessary	3
1.2	The key distinction: fibre cardinality versus observable rank	3
1.3	The two levels of Π : kernel and image	3
1.4	Scope of O24	4
2	The Observable Rank as the Invariant of the Chain	4
2.1	Recapitulation: maximality from O23	4
2.2	Independence of the observable rank from the fibre structure	4
2.3	Residuals and fluctuations as kernel-level phenomena	4
3	The Verticality Lemma	5
3.1	Setup and statement	5
3.2	Case analysis	5
3.3	Remark on the non-classificatory character of the argument	6
4	Main Results	6
4.1	Observable rank stability	6
4.2	Unconditional transfer	6
5	Discussion	7
5.1	What O24 closes and what it does not	7
5.2	Relation to O18: necessary versus sufficient	7
5.3	Vertical versus transversal non-injectivity	7
5.4	Fluctuations as kernel-level residuals	8
6	Conclusion	8
A	Classification of Symmetry Actions: Vertical and Transversal	9

1 Position of the Problem after O22–O23

1.1 What O18 required and why it was sufficient but not necessary

O18 [26] derived the minimal fibre structure of the non-injective projection Π from the parity of the Born–Infeld action $S_{\text{BI}}[\chi]$. Its central result (Theorem 3.3 of O18) states that the fibre of Π above any observable $o \in \mathcal{O}$ contains at least the orbit $\{\chi, -\chi\}$, and that this orbit is the *minimal* fibre under the assumption that Born–Infeld parity is the only global effective symmetry of S_{BI} . Remark 2.6 of O18 acknowledged explicitly that in an enriched framework admitting additional symmetries, the fibre could be larger.

The condition “parity is the only symmetry” was introduced to guarantee minimality of the fibre. It was sufficient for the programme at the time: a minimal fibre $\{\chi, -\chi\}$ ensures that the non-injectivity of Π does not create additional observable directions beyond those already present. However, it is stronger than necessary. The programme does not require small fibres; it requires that additional fibre structure, if present, be *invisible at the level of admissible observables*.

1.2 The key distinction: fibre cardinality versus observable rank

The results of O22 and O23 clarify the precise invariant that controls the chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$. O23 [31] identifies this invariant as

$$\dim_{\mathbb{R}}(\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}) = 3,$$

where $\mathcal{N}_{\text{trl}}$ denotes the admissible neutral traceless sector. This is a property of the *image* of Π , not of its fibres. The integer 3 is the real dimension of the imaginary quaternionic subspace $\text{Im } \mathbb{H}$, identified in O23 as the unique maximal admissible associative non-commutative structure compatible with Born–Infeld parity.

Two configurations that lie in the same fibre of Π are, by definition, mapped to the same observable. Whether the fibre contains two elements or a continuum, the image of Π — and hence the admissible observable structure — is unchanged. The cardinality of the fibres and the rank of the image are therefore independent quantities, and it is the rank that governs the mechanism.

1.3 The two levels of Π : kernel and image

The projection Π admits a natural decomposition into two structurally distinct levels.

The *kernel level* $\ker \Pi$ encodes the microscopic multiplicity: the set of substrate configurations χ that are indistinguishable at the level of admissible observables. A larger kernel means more microscopic configurations projecting to the same effective state. This does not generate new observable directions; it generates projection residuals, i.e. substrate degrees of freedom that are compressed by Π and manifest as fluctuations in the effective description.

The *image level* $\text{Im } \Pi$ encodes the effective observable structure: the set of states that are distinguishable by at least one admissible observable. Its intersection with $\mathcal{N}_{\text{trl}}$ has real rank 3 by O23, and this rank is what enters the threshold condition $\Sigma_c(n_3) = 3$ and, downstream, the transfer $\delta_{\text{pair}} \rightarrow \beta^* \approx 0.126$.

The key structural principle is:

Fluctuations and observables coexist in the same effective space. More residuals do not add new directions to that space; they increase the amplitude of projection noise within the directions already present.

1.4 Scope of O24

The present paper proves the *verticality condition*: every symmetry of S_{BI} compatible with admissibility acts vertically with respect to Π , in the sense that it does not increase the real rank of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$. The proof uses the quaternionic maximality established in O23 as its central ingredient and does not require a global classification of $\mathcal{C}_{\text{eff}}$.

The paper does not address the remaining open task of the programme: the extended numerical campaign at large primes $q \in \{101, 151, 211\}$ across multiple conjugate pairs, required to confirm the shell-alignment prediction and reduce the residual distance $\text{dist}(n_3^{\text{obs}}, \mathbb{Z}) \approx 0.20$ observed at $q = 61$.

2 The Observable Rank as the Invariant of the Chain

2.1 Recapitulation: maximality from O23

We recall the central result of O23 [31].

Theorem 2.1 (O23 Theorem 3.2, Three stable directions from quaternionic maximality). *Let $\Pi: \chi \rightarrow \mathcal{O}$ be a non-injective projection compatible with the Born–Infeld constraint, admitting a Weil realisation on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$. The admissible neutral traceless sector is identified with $\text{Im } \mathbb{H}$. The real rank of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$ is exactly 3. Moreover, this rank is maximal: $\text{Im } \mathbb{H}$ is the unique maximal admissible associative non-commutative structure compatible with Born–Infeld parity, and no admissible realisation of the neutral sector can have real rank greater than 3. Consequently, $\Sigma_c(n_3) = 3$.*

The maximality statement is essential. O23 does not merely establish the *existence* of a quaternionic structure of rank 3; it establishes that no admissible structure compatible with Born–Infeld parity can have a higher rank. This maximality is what makes the verticality argument of Section 3 non-circular.

2.2 Independence of the observable rank from the fibre structure

The following lemma makes the rank–kernel decoupling explicit.

Lemma 2.2 (Rank–kernel decoupling). *The real rank of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$ is independent of the cardinality of $\ker \Pi$.*

Proof. The rank is a property of the image of Π as a linear map; it is determined by the dimension of the span of $\Pi(\mathcal{A})$ within $\mathcal{N}_{\text{trl}}$. Enlarging $\ker \Pi$ — i.e. increasing the number of substrate configurations that project to the same observable — leaves the image $\Pi(\mathcal{A})$ and its intersection with $\mathcal{N}_{\text{trl}}$ unchanged. The rank therefore depends only on the structure of $\text{Im } \Pi$, not on the structure of $\ker \Pi$. $\square$

Remark 2.3. *Lemma 2.2 is linear-algebraic and holds independently of any assumption on the symmetry group G_{BI} . It records the conceptual separation between the two levels of Π identified in Section 1.3: kernel structure governs microscopic multiplicity and fluctuation amplitude; image structure governs the observable mechanism.*

2.3 Residuals and fluctuations as kernel-level phenomena

When the fibre of Π above an observable o contains more than the minimal orbit $\{\chi, -\chi\}$, the additional configurations constitute substrate degrees of freedom that are invisible to admissible observables. These degrees of freedom are not lost; they appear as projection residuals in the effective description.

Formally, let $F_o = \Pi^{-1}(o)$ be the fibre above o . The projection decomposes as

$$\chi = \chi_{\parallel} + \chi_{\perp},$$

where $\chi_{\parallel}$ is the component of χ in the directions visible to Π (i.e. transverse to $\ker \Pi$) and $\chi_{\perp} \in \ker \Pi$ is the residual. The observable $o = \Pi(\chi)$ depends only on $\chi_{\parallel}$. A larger fibre means a larger space of possible $\chi_{\perp}$ values; it does not modify the effective dynamics of $\chi_{\parallel}$.

In particular, the cascade exponent δ_{pair} , the saturation rank n_3 , and the transfer $\delta_{\text{pair}} \rightarrow \beta^*$ all operate at the level of $\chi_{\parallel}$ — the observable sector. Projection residuals $\chi_{\perp}$ appear as fluctuations around the effective dynamics but do not alter the mechanism. Fluctuations and observables coexist in the same effective space; additional residuals increase the noise amplitude within that space — its statistical occupation measure — not its dimensionality. This is consistent with the identification in O14 [22] Section 6.4 of the residual correction $\epsilon(q)$ as the observable imprint of projection residuals, controlled by the phase variance $\text{Var}(\theta)(q)$: more residuals shift the statistical weight within the observable space, leaving its rank unchanged.

3 The Verticality Lemma

3.1 Setup and statement

Let G_{BI} denote the group of symmetries of $S_{\text{BI}}[\chi]$ compatible with Born–Infeld admissibility. A symmetry $g \in G_{\text{BI}}$ is said to be *admissible* if $g(\mathcal{A}) \subset \mathcal{A}$, i.e. it maps the admissible sector to itself. All elements of G_{BI} are admissible by definition; the property is stated explicitly here because it is the key hypothesis used in the proof of Lemma 3.1. A symmetry $g \in G_{\text{BI}}$ is said to act *vertically* with respect to Π if it maps each fibre $\Pi^{-1}(o)$ to itself, i.e. $\Pi(g\chi) = \Pi(\chi)$ for all $\chi \in \mathcal{A}$. A symmetry acts *transversally* if there exists $\chi \in \mathcal{A}$ such that $\Pi(g\chi) \neq \Pi(\chi)$, and moreover $\Pi(g\chi) \notin \text{span}(\Pi(\mathcal{A}))$.

A transversal symmetry therefore generates a new linearly independent direction in $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$: it strictly increases the real rank of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$.

Lemma 3.1 (Verticality of admissible symmetries). *Let $g \in G_{\text{BI}}$ be a symmetry of $S_{\text{BI}}[\chi]$ compatible with Born–Infeld admissibility. Then g acts vertically with respect to Π : it does not increase the real rank of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$.*

Proof. Suppose for contradiction that g acts transversally. Then g generates a new linearly independent direction in $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$.

Since $g \in G_{\text{BI}}$ is admissible, $g(\mathcal{A}) = \mathcal{A}$: any vector generated by the action of g on $\mathcal{A}$ remains in $\mathcal{A}$, and in particular satisfies the Born–Infeld admissibility constraint. Such a direction therefore belongs to the admissible neutral traceless sector characterised in Theorem 2.1.

But Theorem 2.1 (O23) establishes that the admissible neutral traceless sector has *maximal* real rank 3: $\text{Im } \mathbb{H}$ is the unique maximal admissible associative non-commutative structure compatible with Born–Infeld parity. A transversal symmetry generating a 4th independent admissible direction would contradict this maximality. The contradiction shows that g cannot act transversally. Therefore g acts vertically. Thus, no admissible symmetry can increase the dimension of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$. $\square$

3.2 Case analysis

The proof of Lemma 3.1 proceeds by contradiction and applies uniformly to all $g \in G_{\text{BI}}$. For clarity, we spell out how the argument applies to the three natural candidate classes of additional symmetries identified in O18 Remark 2.6 and O23 Section 5.2.

Case 1: Parity $\chi \mapsto -\chi$. This is the symmetry established in O18 [26]. It maps the fibre $\{\chi, -\chi\}$ to itself and is therefore vertical by definition. It does not generate any new direction in $\text{Im } \Pi$.

Case 2: Phase rotation $\chi \mapsto e^{i\theta}\chi$. The substrate field χ is real by construction in the Cosmochrony framework [1]. A $U(1)$ phase rotation is therefore not a symmetry of $S_{\text{BI}}[\chi]$ for $\theta \notin \pi\mathbb{Z}$, and reduces to parity for $\theta = \pi$. This class of symmetries does not arise. Even if the field were complexified, such a symmetry would act within fibres unless it generated new independent admissible neutral traceless directions; but any such directions would be excluded by Lemma 3.1 via the maximality of O23.

Case 3: Topological symmetries. The Topological Invariants paper [4] establishes that admissible configurations are classified by discrete topological invariants (winding number, spin structure) that distinguish configurations with distinct invariant values. Configurations with different topological invariants are therefore distinguishable by Π ; they do not lie in the same fibre and cannot be related by a fibre-preserving symmetry. Any topological symmetry that would identify configurations of different winding number would contradict the discrete classification of [4]. Such symmetries are therefore absent from G_{BI} .

In each case, the conclusion is consistent with Lemma 3.1: no element of G_{BI} acts transversally.

3.3 Remark on the non-classificatory character of the argument

The proof of Lemma 3.1 does not enumerate all possible symmetries of S_{BI} . It instead uses the maximality of O23 as a structural obstruction: any transversal symmetry would produce an admissible direction of rank greater than 3, which is impossible. This argument is therefore independent of a global classification of $\mathcal{C}_{\text{eff}}$ and avoids the topological difficulty identified in Remark 4.2 of [4].

4 Main Results

4.1 Observable rank stability

Theorem 4.1 (Observable rank stability). *Under the Born–Infeld constraint and the Weil realisation on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the real rank of the admissible neutral traceless observable sector $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$ is fixed to 3 as a structural invariant of the admissible sector, independently of the cardinality of the fibres of Π . Consequently, $\Sigma_c(n_3) = 3$ holds unconditionally with respect to the fibre structure of Π . In particular, no admissible enlargement of $\ker \Pi$ can induce any increase in the observable rank.*

Proof. By Lemma 2.2, the rank of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$ is independent of $|\ker \Pi|$. By Lemma 3.1, every $g \in G_{\text{BI}}$ acts vertically and therefore does not increase this rank. By Theorem 2.1 (O23), the rank is rigidly fixed to 3. The combination of these three results gives the claim. The threshold condition $\Sigma_c(n_3) = 3$ is the observable signature of this rank, as established in O23 [31] Section 4.3. $\square$

4.2 Unconditional transfer

Corollary 4.2 (Unconditional transfer $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$). *The chain*

$$c_\chi \longrightarrow \delta_{\text{pair}} \longrightarrow \beta^* = \frac{1}{\delta_{\text{pair}} + \frac{1}{2}} \approx 0.126$$

is independent of the cardinality of the fibres of Π . The conditionality of O18 Theorem 3.3 — which required parity to be the only symmetry of S_{BI} — is resolved: the result holds for any G_{BI} -symmetric non-injective projection, provided the verticality condition of Lemma 3.1 is satisfied. The verticality condition is itself a consequence of the Born–Infeld admissibility constraint and the quaternionic maximality of O23; it requires no additional hypothesis.

Proof. The chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$ was established conditionally in O21 [29] Theorem 5.1, under the shell-alignment conjecture and the condition that $\Sigma_c(n_3) = 3$. O22 [30] proved the shell-alignment conjecture (projection locking forces saturation onto a BFS shell). Theorem 4.1 establishes $\Sigma_c(n_3) = 3$ unconditionally with respect to the fibre structure. The two conditions of O21 Theorem 5.1 are therefore both satisfied unconditionally, and the transfer follows. $\square$

Remark 4.3. Corollary 4.2 does not remove the numerical open direction. The value $\delta_{\text{pair}} \approx 7.44$ and the shell distance $\text{dist}(n_3^{\text{obs}}, \mathbb{Z}) \approx 0.20$ at $q = 61$ are numerical results requiring confirmation at larger primes $q \in \{101, 151, 211\}$ across multiple conjugate pairs. The present result establishes the structural integrity of the chain, not its quantitative precision.

5 Discussion

5.1 What O24 closes and what it does not

The present paper closes the last structural conditionality in the chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$. The three open problems identified across O21–O23 are now resolved at the structural level:

Condition	Origin	Status after O24
Shell-alignment conjecture	O21 Conj. 4.1	Resolved by O22
$\Sigma_c(n_3) = 3$	O21 open problem 2	Resolved by O23
Parity = only symmetry of S_{BI}	O18 Rem. 2.6	Replaced by verticality, proved in O24

What remains open is the extended numerical programme: computing n_3^{obs} for all conjugate pairs with $\delta_{\text{pair}} \in [7.4, 10.6]$ across $q \in \{101, 151, 211\}$ using the canonical pair observable of O19 [27], as described in O22 [30] Section 6.2.

5.2 Relation to O18: necessary versus sufficient

The condition of O18 — parity is the only global symmetry of S_{BI} — was a sufficient condition for the programme at the time it was formulated. It guaranteed that fibres are minimal, which implies in particular that no additional observable directions are created.

The present paper identifies the *necessary* condition: verticality of all admissible symmetries. Verticality is strictly weaker than fibre minimality: a vertical symmetry may enlarge the fibre without changing the image. The replacement of the sufficient condition (O18) by the necessary and sufficient condition (O24) therefore strictly extends the scope of the programme: the chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$ holds not only when fibres are minimal, but whenever additional fibre structure acts vertically.

5.3 Vertical versus transversal non-injectivity

The distinction between vertical and transversal non-injectivity is the structural principle underlying the present paper. It can be stated as a single invariance property:

Observable rank rigidity under admissible non-injectivity. *The real rank of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$ is a protected invariant: $\ker \Pi$ may change freely; $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$ is rigid.*

Vertical non-injectivity means that multiple substrate configurations project to the same observable. This increases the microscopic degeneracy — the number of configurations in each fibre — but leaves the observable sector unchanged. Its physical effect is an increase in the amplitude of projection residuals, manifest as fluctuations in the effective description.

Transversal non-injectivity would mean that Π fails to distinguish configurations that differ in the observable sector. This would collapse independent observable directions and reduce the rank of $\text{Im } \Pi \cap \mathcal{N}_{\text{trl}}$ below 3. Lemma 3.1 shows that this case does not arise for any $g \in G_{\text{BI}}$.

The programme is therefore stable under arbitrary enlargement of the fibres, provided that enlargement is vertical.

5.4 Fluctuations as kernel-level residuals

Section 2.3 identified projection residuals as the physical consequence of vertical non-injectivity. A larger fibre produces a larger space of substrate configurations that are invisible to admissible observables; their contribution appears as fluctuations in the effective description.

Crucially, these fluctuations inhabit the same effective space as the observables. They do not create a new sector; they add noise to the existing one. The cascade exponent δ_{pair} , the saturation rank n_3 , and the threshold condition $\Sigma_c(n_3) = 3$ are all properties of the observable sector and are unaffected by the amplitude of projection noise.

This is consistent with the interpretation developed in O14 [22] Section 6.4, where the residual correction $\epsilon(q)$ is identified as the observable imprint of projection residuals, controlled by the phase variance $\text{Var}(\theta)(q)$. The present result provides the structural justification for this interpretation: projection residuals are precisely the kernel-level degrees of freedom of Π , and their effect on observables is bounded by the rank stability established in Theorem 4.1.

6 Conclusion

The present paper has established that the transfer chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^* \approx 0.126$ holds unconditionally with respect to the fibre structure of the non-injective projection Π . The key result is the Verticality Lemma (Lemma 3.1): every Born–Infeld-admissible symmetry acts vertically with respect to Π , so that no admissible symmetry can enlarge the real rank of the observable sector beyond 3.

The argument is non-classificatory. It does not enumerate all symmetries of S_{BI} or classify the full space $\mathcal{C}_{\text{eff}}$. It uses a single structural obstruction: the quaternionic maximality established in O23 [31], which ensures that no admissible direction of rank greater than 3 can exist.

The conceptual reorientation relative to O18 is:

The condition of O18 required fibres to be small. The present result shows that fibre size is physically irrelevant: non-injectivity affects degeneracy, not structure.

With O22, O23, and the present paper, all structural conditionalities in the O-series chain are resolved. The programme now reduces to a single remaining task: the extended numerical campaign at large primes, which will determine whether the shell-alignment prediction is quantitatively confirmed and reduce the residual uncertainty in δ_{pair} and β^* .

Status update (O25 extension to $q = 601$). That campaign has since been carried out in O25 [32]. An extended pair-capacity campaign up to $q = 601$ confirms that the transfer parameter inferred from the pair observable remains in the expected 0.11–0.13 range, while the underlying *raw* exponent $\delta_{\text{pair}}(q)$ itself descends into the admissible window [7.4, 10.6] as q grows — reaching ≈ 7.6 at $q = 601$ — without requiring the finite-size normalization correction used on the earliest (small- q) datasets. The transfer chain $c_\chi \rightarrow \delta_{\text{pair}} \rightarrow \beta^*$ is thereby confirmed numerically; the exact asymptotic value δ_∞ (and hence the precise value of β^* within that band) remains open, as competing convergence laws are not yet distinguished at $q \leq 601$.

A Classification of Symmetry Actions: Vertical and Transversal

This appendix collects the three natural candidate classes of additional symmetries of S_{BI} and verifies in each case that the action is vertical, supplementing the case analysis of Section 3.2.

A.1 Parity $\chi \mapsto -\chi$. The Born–Infeld action is even in χ : $S_{\text{BI}}[-\chi] = S_{\text{BI}}[\chi]$. Parity therefore belongs to G_{BI} . For any observable $o = \Pi(\chi)$, the definition of the fibre ensures $\Pi(-\chi) = \Pi(\chi) = o$ (since $-\chi$ and χ are identified by the Born–Infeld indiscernability established in O18 [26]). Parity acts vertically.

A.2 Phase rotation $U(1)$. The substrate field $\chi: \text{Heis}_3(\mathbb{Z}/q\mathbb{Z}) \rightarrow \mathbb{R}$ is real-valued by construction [1]. The Born–Infeld kinetic term $F^2 = (D\chi)^2$ is manifestly real and does not admit a $U(1)$ phase symmetry $\chi \mapsto e^{i\theta}\chi$ for $\theta \notin \pi\mathbb{Z}$. This class of symmetries is absent from G_{BI} and requires no further analysis.

A.3 Topological winding symmetries. The Topological Invariants paper [4] establishes that admissible configurations carry discrete topological invariants: charge as a winding invariant (Section 2 of [4]) and spin as a covering invariant (Section 3 of [4]). Two configurations with distinct winding numbers are distinguished by Π : they project to distinct observables. A symmetry that would identify them — mapping a configuration of winding number w to one of winding number $w' \neq w$ — would contradict the discrete classification of [4] and would not be compatible with Born–Infeld admissibility. Such symmetries are absent from G_{BI} .

In all three cases, no transversal action on the admissible observable sector arises. The verticality of G_{BI} established in Lemma 3.1 is therefore consistent with the full catalogue of natural candidate symmetries.

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